A Brief Note on LaTeX

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1 documentclass

```
\documentclass[12pt]{article}
```

Some possible options for the document class:

- article
- report
- book
- letter
- beamer

2 Packages

```
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage{lipsum}
\usepackage{xcolor}
\usepackage{color}
\usepackage{hyperref}
\usepackage{graphicx}
```

- inputenc: Allows the user to specify the input encoding, such as UTF-8.
- amsmath: Provides various features to facilitate writing math formulas.
- amsfonts: Provides additional math fonts.
- amssymb: Provides additional math symbols.
- lipsum: Generates dummy text (Lorem Ipsum).
- xcolor: Provides support for colored text.
- color: Provides support for colored text (older package, often replaced by xcolor).

- hyperref: Enables the creation of hyperlinks.
- graphicx: Allows the inclusion of graphics.

3 Meta information

3.1 Title

```
\title{A Brief Note on \LaTeX}
```

3.2 Author

```
\author{Mehmet Hakan Satman}
```

3.3 Date

```
\date{\today}
```

3.4 Making Title

```
\maketitle
```

4 Table of Contents

```
\tableofcontents
```

5 Paragraphs

This is the first paragraph. It contains some text to demonstrate how paragraphs are created in LaTeX.

This is the second paragraph. Notice the space between this paragraph and the previous one, which indicates a new paragraph has started.

This is the third paragraph. It follows immediately after the second paragraph without any additional vertical space.

```
This is the first paragraph. It contains some text to
demonstrate how paragraphs are created in \LaTeX.

This is the second paragraph. Notice the space between
 this paragraph and the previous one, which indicates
  a new paragraph has started.

\noindent This is the third paragraph. It follows immediately
 after the second paragraph without any additional vertical space.
```

6 Centering

This text is centered in the document.

```
\begin{center}
This text is centered in the document.
\end{center}
```

7 Quotes

Inline Quote: "This is an inline quote in LaTeX."

```
``This is an inline quote in \LaTeX.''
```

8 Bold, italic, and underline Text

Bold Text: **This text is bold.**

```
\textbf{This text is bold.}
```

Italic Text: *This text is italic.*

```
\textit{This text is italic.}
```

Underline Text: This text is underlined.

```
\underline{This text is underlined.}
```

Bold and Italic Text: ***This text is bold and italic.***

```
\textbf{\textit{This text is bold and italic.}}
```

Bold and Underline Text: **This text is bold and underlined.**

```
\textbf{\underline{This text is bold and underlined.}}
```

Bold, italic, and Underline Text: ***This text is bold, italic, and underlined.***

```
\textbf{\textit{\underline{This text is bold, italic, and underlined.}}}
```

Script Text: This text is in small capitals.

```
\textsc{This text is in small capitals.}
```

Computer code: `This text is in typewriter font.`

```
\texttt{This text is in typewriter font.}
```

9 Colored Text

9.1 xcolor

Required package: `\usepackage{xcolor}`

Red : This text is red.

```
\textcolor{red}{This text is red.}
```

Blue: This text is blue.

```
\textcolor{blue}{This text is blue.}
```

Green: This text is green.

```
\textcolor{green}{This text is green.}
```

Orange: This text is orange.

```
\textcolor{orange}{This text is orange.}
```

9.2 color

Required package: `\usepackage{color}`

Red : This text is red.

```
\textcolor{red}{This text is red.}
```

Yellow background: This text has a yellow background.

```
\colorbox{yellow}{This text has a yellow background.}
```

Yellow background with red border:
This text has a yellow background with a red border.

```
\fcolorbox{red}{yellow}{This text has a yellow background with a red border.}
```

9.3 tcolorbox

Required package: `\usepackage{tcolorbox}`

tcolorbox with default settings.

```
\begin{tcolorbox}
  tcolorbox with default settings.
\end{tcolorbox}
```

Example

This is a tcolorbox.

```
\begin{tcolorbox}[colback=green,colframe=blue,title=Example]
\textbf{This is a tcolorbox.}
\end{tcolorbox}
```

Another Example

This is another tcolorbox.

```
\begin{tcolorbox}[colback=white,colframe=black,title=Another Example]
\textbf{This is another tcolorbox.}
\end{tcolorbox}
```

10 URLs

Required package: `\usepackage{hyperref}`

https://www.example.com

```
\url{https://www.example.com}
```

11 Sectioning Commands

Section:

```
\section{Section Title}
```

Subsection:

```
\subsection{Subsection Title}
```

Subsubsection:

```
\subsubsection{Subsubsection Title}
```

12 Mathematical Expressions

$$E = mc^2$$

```
$$
E = mc^2
$$
```

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

```
$$
\int_a^b f(x) \, dx = F(b) - F(a)
$$
```

$$\Delta = b^2 - 4ac$$

```
$$
\Delta = b^2 - 4ac
$$
```

$$\int_0^\infty \frac{1}{\theta} e^{-\frac{x}{\theta}} \, dx = 1$$

```
$$
\int_0^\infty \frac{1}{\theta} e^{-\frac{x}{\theta}} \, dx = 1
$$
```

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$$

```
$$
p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0
$$
```

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

```
$$
f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n
$$
```

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

```
$$
f(x) = \begin{cases}
      x^2 & x \geq 0 \\
      -x & x < 0
   \end{cases}
$$
```

$$sign(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

```
$$
sign(x) = \begin{cases}
      1 & x > 0 \\
      0 & x = 0 \\
      -1 & x < 0
   \end{cases}
$$
```

$$sign(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

```
$$
abs(x) = \begin{cases}
      x & x \geq 0 \\
      -x & x < 0
   \end{cases}
$$
```

$$abs(x) = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

```
$$
A = \begin{bmatrix}
1 & 2 & 4 \\
0 & 1 & 3 \\
0 & 0 & 1
\end{bmatrix}
$$
```

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

```
$$
\max z = x + 2y
$$
Subject to:
$$
\begin{aligned}
2x + 3y &\leq 52 \\
x + y &\geq 2 \\
x &\geq 0, y \geq 0
\end{aligned}
$$
```

$$\max z = x + 2y$$

Subject to:

$$\begin{aligned} 2x + 3y &\leq 52 \\ x + y &\geq 2 \\ x &\geq 0, y \geq 0 \end{aligned} \tag{1}$$

```
\begin{equation}
\begin{split}
    \max z = & x + 2y \\
    \text{Subject to:} &  \\
    & 2x + 3y \leq 52 \\
    & x + y \ge 2 \\
    & x \geq 0, y \geq 0 \\
\end{split}
\end{equation}
```

Suppose the quadratic equation is given by:

$$f(x) = ax^2 + bx + c.$$

so $\Delta = b^2 - 4ac$.

```
$$
f(x) = ax^2 + bx + c.
$$

\noindent so $\Delta = b^2 - 4ac$.
```

The regression equation is given by:

$$y = \beta_0 + \beta_1 x + \epsilon$$

where y is the dependent variable, x is the independent variable, β_0 and β_1 are the regression coefficients, and ϵ is the error term.

```
$$
y = \beta_0 + \beta_1 x + \epsilon
$$

\noindent where $y$ is the dependent variable, $x$ is the
independent variable, $\beta_0$ and $\beta_1$
are the regression coefficients, and $\epsilon$
is the error term.
```

Binomial coefficient n of k is given by:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

```
$$
\begin{pmatrix}
    n \\
    k
\end{pmatrix} = \frac{n!}{k!(n-k)!}
$$
```

$$\binom{n}{x} = \frac{n!}{k!(n-k)!}$$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

is the probability mass function of the Binomial distribution.

```
$$
P(X = x) = \begin{pmatrix} n \\ x \end{pmatrix} p^x (1-p)^{n-x}
$$

\noindent is the probability mass function of the Binomial distribution.
```

$$f(x, \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

is the probability mass function of the Poisson distribution.

```
$$
f(x, \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}
$$

\noindent is the probability mass function of the Poisson distribution.
```

12.1 Common Mathematical Symbols

Common mathematical symbols used in LaTeXdocuments are listed in Table 1 and Table 2.

13 Lorem Ipsum

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida

Order	Symbol	Code
1	α	`\alpha`
2	β	`\beta`
3	γ	`\gamma`
4	δ	`\delta`
5	Δ	`\Delta`
6	ϵ	`\epsilon`
7	ε	`\varepsilon`
8	γ	`\gamma`
9	Γ	`\Gamma`
10	ρ	`\rho`
11	η	`\eta`
12	ζ	`\zeta`
13	θ	`\theta`
14	λ	`\lambda`
15	μ	`\mu`
16	σ	`\sigma`
17	π	`\pi`
18	Σ	`\Sigma`
19	$\int$	`\int`
20	$\sum$	`\sum`
21	$\frac{a}{b}$	`\frac{a}{b}`
22	$\sqrt{x}$	`\sqrt{x}`
23	$\sqrt[n]{x}$	`\sqrt[n]{x}`
24	∞	`\infty`
25	x_i	`x_i`
26	x^n	`x^n`
27	x_i^n	`x_{i}^{n}`

Table 1: Common Mathematical Symbols in LaTeX

mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices

Order	Symbol	Code
1	$\pm$	\pm
2	$\mp$	\mp
3	$\times$	\times
4	$\div$	\div
5	$\cdot$	\cdot
6	$\leq$	\leq
7	$\geq$	\geq
8	$\neq$	\neq
9	$\approx$	\approx
10	$\forall$	\forall
11	$\exists$	\exists
12	$\in$	\in
13	$\notin$	\notin
14	$\subset$	\subset
15	$\subseteq$	\subseteq
16	$\cup$	\cup
17	$\cap$	\cap
18	$\setminus$	\setminus
19	$\emptyset$	\emptyset
20	$\mathcal{R}$	\mathcal{R}
21	$\mathbb{R}$	\mathbb{R}
22	$\mathbb{N}$	\mathbb{N}
23	$\mathbb{Z}$	\mathbb{Z}
24	$\mathbb{Q}$	\mathbb{Q}
25	$\mathbb{C}$	\mathbb{C}
26	$\ldots$	\ldots

Table 2: Common Mathematical Symbols in LaTeX(Continued)

bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet

magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

```
\lipsum[1-2]
```

14 Tables

14.1 Tabular Environment

A	B	C
1	2	3
4	5	6

```
\begin{tabular}{|c|c|c|}
\hline
A & B & C \\
\hline
1 & 2 & 3 \\
4 & 5 & 6 \\
\hline
\end{tabular}
```

14.2 Tables with Caption

X	Y	Z
7	8	9
10	11	12

Table 3: A Simple Table

```
\begin{table}[h!]
\centering
\begin{tabular}{|c|c|c|}
\hline
```

```
X & Y & Z \\
\hline
7 & 8 & 9 \\
10 & 11 & 12 \\
\hline
\end{tabular}
\caption{A Simple Table}
\end{table}
```

14.3 Table placement options

- h: Place the table here, approximately at the same point it occurs in the source text.
- t: Position the table at the top of the page.
- b: Position the table at the bottom of the page.
- p: Put the table on a special page (in a separate page).
- !: Override internal parameters LaTeX uses for determining "good" float positions.
- H: Places the float at precisely the location in the LaTeX code. Requires the 'float' package.

14.3.1 Usage

```
% Put table on top
\begin{table}[t]

% Put table on bottom
\begin{table}[b]

% Put table on a special page
\begin{table}[p]
```

15 Listings

15.1 Itemize Environment

- First item
- Second item
- Third item

```
\begin{itemize}
    \item First item
    \item Second item
    \item Third item
\end{itemize}
```

15.2 Enumerate Environment

1. First item
2. Second item
3. Third item

```
\begin{enumerate}
    \item First item
    \item Second item
    \item Third item
\end{enumerate}
```

15.3 description

LaTeX: A typesetting system.

HTML: A markup language.

Word: A word processing software.

Excel: A spreadsheet software.

beamer: A presentation package for creating slides in LaTeX.

```
\begin{description}
  \item[\LaTeX{}:] A typesetting system.
  \item[HTML:] A markup language.
  \item[Word:] A word processing software.
  \item[Excel:] A spreadsheet software.
  \item[beamer:] A presentation package for creating slides in \LaTeX{}.
\end{description}
```

15.4 Text in boxes

This is a text box in LaTeX. You can use it to highlight important information or to create a visual separation from the surrounding text.

```
\begin{center}
\framebox{
    \begin{minipage}{0.8\textwidth}
    This is a text box in \LaTeX. You can use it to highlight
    important information or to create a visual
    separation from the surrounding text.
    \end{minipage}
}
\end{center}
```

16 Referencing

16.1 Equation Referencing

$$\Delta = b^2 - 4ac \tag{2}$$

```
\begin{equation}
\Delta = b^2 - 4ac
\label{eq:delta}
\end{equation}
```

Referencing: Equation 2 shows the discriminant of a quadratic equation.

```
Referencing: Equation \ref{eq:delta} shows the discriminant
of a quadratic equation.
```

16.2 Table Referencing

A	B	C
1	2	3
4	5	6

Table 4: A Simple Table

```
\begin{table}[h!]
\centering
\begin{tabular}{|c|c|c|}
\hline
A & B & C \\
\hline
1 & 2 & 3 \\
4 & 5 & 6 \\
\hline
\end{tabular}
\caption{A Simple Table}
\label{tab:simple}
\end{table}
```

Table 4 shows a simple table with three columns and two rows of data.

```
Table \ref{tab:simple} shows a simple table with
three columns and two rows of data.
```

17 Figures

17.1 Including Figures

```
\begin{figure}[h!]
\centering
\includegraphics[width=0.5\textwidth]{iulogo.png}
\caption{An Example Figure}
\label{fig:example}
\end{figure}
```



Figure 1: An Example Figure

Referencing: Figure 1 shows an example figure.

```
Referencing: Figure \ref{fig:example} shows an example figure.
```

18 Presentations

Required package: `beamer`

Example:

```
\documentclass{beamer}
\begin{document}
\begin{frame}
\frametitle{Sample Slide}
This is a sample slide in a Beamer presentation.
\end{frame}
\end{document}
```

Changing the theme:

```
\documentclass{beamer}
\usetheme{Madrid}
\begin{document}
```

```
\begin{frame}
\frametitle{Sample Slide with Theme}
This is a sample slide in a Beamer presentation with the Madrid theme.
\end{frame}
\end{document}
```

Some possible themes for Beamer presentations include:

- Madrid
- Berlin
- Copenhagen
- CambridgeUS
- Frankfurt

19 Expression trees with Forest package

Example:

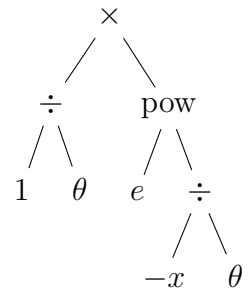
```
\begin{forest}
  [+
    [a]
    [b]
  ]
\end{forest}
```

Rendered output:



19.1 The Exponential Distribution

The function: $f(x;\theta) = \frac{1}{\theta}e^{-\frac{x}{\theta}}$

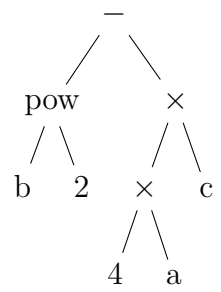


The code:

```
\begin{forest}
  [$\times$
    [$\div$
      [1]
      [$\theta$]
    ]
    [pow
      [$e$]
      [$\div$
        [$-x$]
        [$\theta$]
      ]
    ]
]
\end{forest}
```

19.2 Delta!

The expression: $\Delta = b^2 - 4ac$.



The code:

```
\begin{forest}
  [$-$
    [pow
      [b]
      [2]
    ]
    [$\times$
        [$\times$
        [4]
        [a]
        ]
        [c]
    ]
  ]
\end{forest}
```